Sequential Harm Audits Under Selective Reporting

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Abstract

Incident reports about an agent intervention may arrive more often when the intervention causes harm. A sequential test can then raise false alarms even when each underlying case is distinct. We derive a sharp adjustment for a bounded ratio of reporting probabilities and prove anytime validity with adaptive case histories. We also give a rule for using a time-uniform upper bound learned from calibration while preserving past report evidence. Exact finite experiments compare the adjustments with naive monitoring and independent audits. A paired construction shows which changes in underlying harm remain unidentified from reports. An independent audit top-up gives a sharp reduction in the reporting-ratio bound when cases are deduplicated. When the full case count is independently observed, a second test is valid even with unbounded reporting bias. Keeping audit and report outcomes separate also gives an exact log-optimal factor for a specified alternative. We compare when these tests detect harm.

Summary and scope

Question. How can an operator decide when to intervene if users report harms selectively?

Finding. A conditional reporting-ratio cap gives an anytime harm test. An upper bound learned with simultaneous calibration coverage adds only its coverage error to the crossing bound. A pooled reporting ratio may fail under predictable context shifts.

Evidence and scope. The guarantees are proved and checked with exact crossing calculations and observationally identical counterexamples. The operator needs a bound for the cases on which the monitor bets and an independent route for individual appeals.

1 The harm rate before a case is reported

Suppose a scheduling agent changes appointments on behalf of its users. Some users report disruptions and others never contact the service. The operator wants to intervene if the underlying harm rate exceeds a reference value p_0, even when evidence arrives slowly enough to make a fixed sample size impractical. Merging duplicate reports is necessary for this question. It still leaves the selection of which cases are reported.

Let $X_t \in \{0, 1\}$ denote harm in case t and $A_t \in \{0, 1\}$ indicate whether its outcome enters the audit. The observed outcome is either omitted, admitted clean, or admitted harmful. The history $\mathcal{F}_{t-1}$ includes information available when the monitor sets its next betting factor. We consider the conditional null $\Pr(X_t = 1 \mid \mathcal{F}_{t-1}) \leq p_0$. The selection model permits different admission probabilities for harmful and clean cases, with ratio at most ρ at each history.

The main result separates two choices an operator can make. The reference rate p_0 expresses a tolerance for underlying harm. The reporting ratio ρ reflects the audit's observation process. This

response probability ratio is a sensitivity parameter for outcome-dependent observation. Smith and VanderWeele [1] bound causal risk-ratio distortion under selection. Here we use a conditional reporting ratio to construct a sequential test, with a nonnegative supermartingale as in time-uniform inference [2].

2 A sharp conditional reporting bound

Theorem 1. *Let $0 < p_0 < 1$ and $1 \le \rho < \infty$. At every history let $a_x = \Pr(A_t = 1 \mid X_t = x, \mathcal{F}_{t-1})$, with $a_1 \le \rho a_0$. Set*

$$q = \frac{\rho p_0}{1 - p_0 + \rho p_0}.$$

When $\Pr(A_t = 1 \mid \mathcal{F}_{t-1}) > 0$, the admitted harm probability is at most q under the conditional null. Let b_t be any predictable value in $[q, 1]$. Start with $M_0 = 1$ and multiply by 1 on omission, by b_t/q on admitted harm, and by $(1 - b_t)/(1 - q)$ on admitted clean outcomes. For any $0 < \alpha < 1$,

$$\Pr(\exists t \ge 0 : \ M_t \ge 1/\alpha) \le \alpha.$$

The bound q on admitted harm is sharp.

Proof. Fix a history and write $p = \Pr(X_t = 1 \mid \mathcal{F}_{t-1})$. At histories with positive admission, Bayes' rule gives

$$\Pr(X_t = 1 \mid A_t = 1, \mathcal{F}_{t-1}) = \frac{p a_1}{p a_1 + (1 - p) a_0} \le \frac{\rho p}{1 - p + \rho p} \le q.$$

The odds of harm have been multiplied by at most ρ. Positive admission under the null implies $a_0 > 0$, since $a_0 = 0$ forces $a_1 = 0$ by assumption. Null conditional events can be assigned arbitrary consistent versions of a_x.

Write r for the admitted harm probability. The mean factor on admission is

$$r \, \frac{b_t}{q} + (1 - r) \frac{1 - b_t}{1 - q} = 1 + (r - q) \frac{b_t - q}{q(1 - q)} \le 1.$$

Omissions contribute factor one, so M_t is a nonnegative supermartingale. Stop it at its first crossing or a fixed finite horizon. Its expected stopped value is at most one, while on crossing its value is at least $1/\alpha$. Letting the horizon increase gives the result. For sharpness, take $p = p_0$, $a_1 = 1$ and $a_0 = 1/\rho$. □

The factor can use a history-dependent bet without changing the argument. A large ρ brings q close to one and makes successive harmful reports weak evidence. A process that learns how strongly to bet may improve power, but it still needs an externally justified upper bound on selection. The rate q can also vary predictably with the case class when the reference rate and reporting bound are specified conditionally.

3 Reducing selection bias with independent audit draws

Let R indicate an adjudicated spontaneous report. Before using any current-case evidence, choose an audit probability s from the past history. Select the audit with probability s independently of both the harm and the report, conditional on that history. Admit a case if either route provides the outcome. If both routes apply, record it once. An audit assignment can reuse an already adjudicated outcome. This requires comparable outcome definitions on both routes.

Proposition 1. *If spontaneous reporting satisfies $a_1 \le \rho a_0$, the combined admission probabilities satisfy the sharp ratio bound*

$$\rho_{\text{pool}} = \frac{\rho}{1 + s(\rho - 1)}.$$

With no finite spontaneous reporting bound and $s > 0$, the sharp combined ratio bound is $1/s$.

Proof. The combined probability for outcome x is $s + (1-s)a_x$. For fixed a_0, the largest numerator uses $a_1 = \min(1, \rho a_0)$. Up to $a_0 = 1/\rho$, the ratio is nondecreasing in a_0. Beyond that point the numerator is one, and the ratio is nonincreasing. The maximum is attained at $(a_0, a_1) = (1/\rho, 1)$. With unrestricted reporting the largest numerator is one and the smallest denominator is s, attained by $(0, 1)$. □

Use ρ_{pool} in the conditional harm bound at each history. For $\rho > 1$, a target combined ratio $r \in [1, \rho]$ requires $s \ge (\rho/r - 1)/(\rho - 1)$. The expected number of additional adjudications is $s \Pr(R = 0)$ per opportunity if already reported cases have a usable label. A draw concentrated on already suspected harms needs a different observation model. Combining two routes for one case does not create two independent outcomes.

The verifier compared 4,514 grid ratios, with a sharp case for each of 36 settings of ρ and s. For the existing ratio-three example, set $s = 1/4$, giving a combined ratio of two. Over 48 opportunities with threshold eight, use the same harm alternative 4/5 to set a fixed bet for each observation rule. The resulting crossing probabilities are

Observation rule	Null at 1/2	Alternative at 4/5
Spontaneous reports	0.0888	0.9013
Deduplicated pool	0.0958	0.9543
Independent assignments only	0.0606	0.6469

The pooled and assignment-only rules use the same random assignments. The pool expects four extra adjudications on unreported cases under the null and 8/5 under the alternative. These are model-based costs and crossing probabilities for the stipulated reporting laws.

4 Using all enumerated opportunities

Suppose the monitor observes every eligible case even when neither route provides a label. Let $Y = 1$ mean that an adjudicated harmful outcome entered the audit and let $Y = 0$ for all other cases. Since $Y \le X$, the conditional null also bounds $\Pr(Y = 1)$ by p_0. This gives a valid betting factor for any predictable $d \in [p_0, 1]$, using d/p_0 on $Y = 1$ and $(1 - d)/(1 - p_0)$ otherwise. It requires a complete case count and no false harmful labels. It uses omissions as evidence about the detected harm rate. If reporting suppresses so many harms that this rate remains below p_0, it cannot provide the intended power.

The monitor can also retain the route by which it obtained each label. With audit probability s and spontaneous reporting probabilities r_0, r_1, there are five outcomes. They are audited harm, audited clean, unaudited reported harm, unaudited reported clean, and omission. Their probabilities are

$$\big(sp,\ s(1-p),\ (1-s)pr_1,\ (1-s)(1-p)r_0,\ (1-s)[p(1-r_1) + (1-p)(1-r_0)]\big).$$

An independently assigned audit supplies a usable label in this model. Count each case on exactly one route.

Proposition 2. *For known s and finite $\rho \geq 1$, a nonnegative five-outcome factor has conditional expectation at most one for every null law with $p \leq p_0$ and $r_1 \leq \rho r_0$ exactly when it passes that check at $p \in \{0, p_0\}$ and*

$$(r_0, r_1) \in \{(0,0), (1,0), (1,1), (1/\rho, 1)\}.$$

Proof. For fixed reporting probabilities the expectation is affine in p, so its maximum is at one endpoint. At each endpoint it is linear on the reporting polygon $0 \leq r_0, r_1 \leq 1$, $r_1 \leq \rho r_0$. The listed points are the vertices, with a repeat when $\rho = 1$. Every vertex law is allowed by the null. □

For example, use the ordinary harm threshold in the assigned branch and the report-adjusted threshold in the unassigned branch. The five-outcome check certifies this rule and allows other factors that use the route information. The verifier checked 1,152 conditional expectations for denominator, pooled and route-specific factors at the extreme null laws.

For the denominator and pooled rules, a fixed weighted average of their wealth processes is also a nonnegative supermartingale. The same anytime threshold applies to that average. It shares evidence across the two rules without treating their outcomes as independent. The following exact comparison uses equal weights, 24 opportunities, threshold eight, $s = 1/4$, and $\rho = 3$. The denominator bet is $d = 4/5$ and the pooled bet is $8/9$. The null uses $p = 1/2$ with $(r_0, r_1) = (1/3, 1)$. Both alternatives use $p = 4/5$, with reporting pairs $(1/3, 1)$ and $(1/10, 3/10)$ respectively.

Case law	Denominator	Pooled	Equal mixture
Null	0.0826	0.0730	0.0788
All harms reported	0.8978	0.7945	0.8999
Sparse reports	0.0144	0.2970	0.1545

The mixture gives up power relative to the pooled rule in the sparse fixture. A service can choose the mix before evaluating the live stream, using separate evidence about how complete its case and reporting records are. Exact path enumeration agreed with the crossing recursion on nine short runs.

5 Exact finite comparison

We computed first-crossing probabilities for 48 opportunities, threshold eight, and independent opportunities within each fixture. The naive process uses $q = 1/2$ and $b = 3/4$. The adjusted process uses $q = 3/4$ and $b = 7/8$. The independent audit selects each case with probability $2/3$ and applies the $1/2$ reference. In the primary null the expected sample size matches that of admitting all harmful cases and one third of clean cases.

Selection	Monitor	$p = 1/2$	$p = 4/5$
Ratio 3	Naive	0.8809	1.0000
Ratio 3	Adjusted	0.0620	0.9188
Ratio 4	Adjusted using 3	0.1528	0.9672
Independent audit	1/2 reference	0.0919	0.9659

Table entries are rounded from exact rationals. The algorithm propagates surviving probability mass over pairs (h, c) of admitted harmful and clean counts. On each transition it removes mass when $(b/q)^h((1-b)/(1-q))^c \geq 8$. The omitted transition leaves the pair unchanged. The verifier checks conservation of mass and compares a seven-step case with independent enumeration of all

3^7 paths. The ratio-four condition intentionally violates the assumed bound. Its null crossing probability exceeds $1/8$ in this finite experiment. Under the specified alternative and the primary selection model, the corrected monitor crosses in about 92% of cases. This power belongs to that fixture.

6 An identification limit

Proposition 3. *A known upper bound $\rho = 3$ on the response probability ratio does not generally identify whether the underlying harm rate exceeds $1/2$. This remains true when harms are at least as likely to be reported as clean outcomes.*

Proof. In the null world take $(p, a_1, a_0) = (1/2, 3/5, 3/10)$. In the alternative take $(3/5, 1/2, 3/8)$. Each world has an admitted harm with probability $3/10$, an admitted clean outcome with probability $3/20$, and an omission with probability $11/20$. Both satisfy $1 \le a_1/a_0 \le 3$. Repeating each world independently gives identical laws for all observed histories. Consequently any test based on those histories has identical crossing probabilities in the two worlds. □

7 Learning an upper bound without restarting the monitor

Suppose a fixed $\rho_\star$ bounds the conditional reporting ratio at all histories, but its value is unknown. Write $q(\rho) = \rho p_0/(1 - p_0 + \rho p_0)$ and $q(+\infty) = 1$. Use a predictable coefficient $\lambda_t \in [0, 1]$, chosen once at each time for all candidate ratios. For a candidate ρ, let $E_t(\rho)$ be the product of factors

$$1 \quad \text{on omission,} \qquad 1 + \lambda_t(X_t - q(\rho)) \quad \text{on admission.}$$

These factors are nonnegative and decrease with ρ on every realized path. For $\rho_\star$, the conditional mean factor on admission is at most one, so $E_t(\rho_\star)$ is a nonnegative supermartingale under the null. Keep the historical coefficients fixed when evaluating a new candidate ratio.

Theorem 2. *Let calibration produce upper bounds $U_t \ge 1$ with $\Pr(\exists t : U_t < \rho_\star) \le \beta$. Then stopping when $E_t(U_t) \ge 1/\alpha$ has null crossing probability at most $\alpha + \beta$.*

Proof. On the event of simultaneous calibration coverage, pathwise monotonicity gives $E_t(U_t) \le E_t(\rho_\star)$ for all t. On that event, crossing by the calibrated process implies crossing by the fixed-ratio process, which has probability at most α. Add the probability at most β that the upper bound ever fails. The argument needs no independence between the two processes when the conditional null and ratio bound hold in the monitor's full pre-case filtration. The guarantee comes from comparison with $E_t(\rho_\star)$. The plug-in process itself need not be a supermartingale. □

For example, with fixed reporting probabilities a_0, a_1, representative independent audit samples within each outcome stratum can give a simultaneous upper bound. After n audited cases in a stratum, use one-sided Hoeffding radius

$$d_n = \sqrt{\frac{\log(2n(n+1)/\beta)}{2n}}.$$

Take the lower endpoint for a_0 and the upper endpoint for a_1, clipped to the unit interval. Their respective error probabilities are each at most $\beta/(2n(n+1))$. Summing over both strata and every n gives total probability at most β. Divide the upper endpoint by the lower one, flooring the result at one and using $+\infty$ when the lower endpoint is zero or a stratum is missing. The resulting upper bound can be reevaluated as audits arrive. A narrower interval does not justify changing the coefficients of earlier bets.

A pooled calibration can fail. Let two observed contexts be equally common, with harm probability 1/2 in both. Their (a_0, a_1) pairs are $(1/4, 1)$ and $(3/4, 0)$. Pooled reporting probabilities are equal, yet the first context has ratio four. A monitor that bets only in that context, uses ratio one, and sets $\lambda_t = 1$ has conditional mean factor 19/16 under the null there. An upper bound on a pooled reporting ratio cannot replace a bound at histories where the monitor bets.

8 An exact calibration experiment

In a finite two-point model, the clean reporting probability is 1/4 and the true ratio is one or three. Before opportunity four, use $U_t = 3$. At that opportunity reveal the reporting indicators from two independently adjudicated harmful cases. Lower U_t to one for the remaining decisions only if both were unreported. At true ratio three, this happens with probability 1/16, which bounds simultaneous undercoverage in this experiment. Calibration therefore spends two extra known-harm audits. We use coefficient one and threshold eight over 40 opportunities. The theorem's error allowance is $1/8 + 1/16 = 3/16$.

True setting	Fixed upper three	Calibrated	Known ratio	Naive ratio one
Null, ratio one	0.0005	0.0232	0.0408	0.0408
Null, ratio three	0.0797	0.1205	0.0797	0.7324
Alternative, ratio one	0.0608	0.3241	0.5289	0.5289
Alternative, ratio three	0.7605	0.7754	0.7605	0.9997

The alternative is $p = 4/5$. Entries are rounded exact first-crossing probabilities. At true ratio one, calibration recovers part of the power lost by always using three. At ratio three, occasional undercoverage raises the null crossing probability relative to knowing the true ratio. These operating points belong to the declared two-point model. The calculation agrees with 8,748 six-step path evaluations across twelve oracle settings. The full receipt retains the exact values.

9 A factor tuned to sparse reporting

A monitor that retains the observation route can target a specified alternative. Let Q be its strictly positive five-outcome law, and let P^* be a strictly positive mixture of admissible null laws. Consider the candidate factor $e^*(i) = Q_i/P^*_i$. If its mean is at most one at each null vertex, it is valid and has the largest expected log factor under Q among all valid factors. This is the finite form of the reverse information projection construction of Larsson, Ramdas, and Ruf [3]. Indeed, any valid e has $\sum_i P^*_i e(i) \le 1$, so Jensen's inequality gives

$$E_Q[\log(e/e^*)] \le \log\left(\sum_i P^*_i e(i)\right) \le 0.$$

Here is a family where the factor is explicit. Fix $p_0 < p_1 < 1$ and an audit probability $0 < s < 1$. In the alternative set $r_1 = \rho r_0$ and let m be its total spontaneous reporting probability. Assume $0 < m < 1$ and $m \le p_0 + (1 - p_0)/\rho$. For the null candidate use p_0 with the same m and the same reporting ratio. Both laws are feasible because their harm-report probability is obtained from $r_1 = \rho m/(1 - p + \rho p) \le 1$. Set $q_j = \rho p_j/(1 - p_j + \rho p_j)$. The tagged factor is

$$\left(\frac{p_1}{p_0},\ \frac{1-p_1}{1-p_0},\ \frac{q_1}{q_0},\ \frac{1-q_1}{1-q_0},\ 1\right).$$

Under the null, the harm probability is at most p_0 on the audit route and at most q_0 conditional on an unaudited report. Both conditional mean factors are at most one. Omissions have factor one. This proves validity and, by the preceding argument, log optimality for the stated alternative.

For the earlier sparse fixture the factor is $(8/5, 2/5, 16/13, 4/13, 1)$. The 24-opportunity crossing probability is 0.4956, compared with 0.2970 for the pooled factor and 0.0144 for the denominator factor. In the dense alternative, using the sparse factor lowers the crossing probability to 0.7634 from the denominator's 0.8978. For that dense alternative the denominator factor is itself log optimal, by taking P^* with the dense reporting parameters and $p = p_0$. The sparse factor crossed with probability 0.0746 and 0.0514 in the dense and sparse null fixtures. All use the same threshold of eight. The verifier checked 304 vertex constraints and reproduced 12 short crossings by full path enumeration. Log optimality concerns expected growth under the declared alternative. The finite-horizon comparisons concern the individual simulated laws.

A monitor can use past or separate evidence to choose a factor before the current case. Fixing the factor for sparse reports has a cost when reporting is dense. Retaining route labels gives a choice between these factors while preserving the null test as long as each chosen factor satisfies the conditional constraint.

10 Using the audit to correct an agent

An operator can use independent sampling to investigate silent harm or use a justified response ratio to interpret existing reports. The first choice costs review time. The second can reduce that cost where the reporting channel is well characterized. A separate appeal route remains necessary because a statistical alarm identifies a pattern of outcomes rather than adjudicating a specific person's case. A practical first deployment would be a shadow audit on paired notification or scheduling interventions, with an independent random sample used to estimate reporting differences. Stratify by how easily users can report. The prospective test should compare alarm timing, missed harms, and review cost while preserving an independently observed denominator. Unverified case identities, delayed outcome adjudication, and selection that shifts after an alarm require additional controls.

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References

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